Methods

Speech flexibility

- Flexibility of each phoneme type was estimated as its conditional entropy through mixture density networks (MDN; cf. Richmond 2006; Zen & Senior 2014).
- MDN predicts a K-mixture (K=5) probability density distribution based on the input conditions.
- In this study:
  - P(observed articulatory matrix | acoustic matrix)
  - MDNs were first trained on the train sets & used to estimate entropies for the test sets.

Richmond, K. (2006). A trajectory mixture density network for the acoustic-articulatory inversion mapping. In INTERSPEECH 2006 – ICSLP Ninth International Conference on Spoken Language Processing (Paper 1790-Mon3WeS.4). International Speech Communication Association. Zen, H., & Senior, A. (2014). Deep mixture density networks for acoustic modeling in statistical parametric speech synthesis. In Proceedings of the IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP).

Methods

Speech flexibility–MDN prediction

- A /k/ token produced by Sub013 in AE-MRI-annotated (PCA used for illustration purposes).